



tecnun

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● C++ Laboratory Lessons

Lesson N° 1

C++
PARA INGENIEROS SUPERIORES

Computer Programming II

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1. Introduction

Computer Programming II begins with some easy examples, dealing with C++ programming, which will help you to learn some basic concepts.

In order to complete the *Exercises* in this lesson, you must learn on *Visual Studio programming environment*, and remember to always use the *Debugger*.

1.1 Exercise 1: My first program in C++

In order for your introduction to this C++ Programming course to be a success, we have prepared a simple welcoming program.

```
/* File Exer1.cpp */
// welcome program - Tecnun2025
#include <iostream>
using namespace std;    //for Visual Studio 2019 or superior

void main (void) {
    char name[30];        //variables declaration
    int age;

    cout << "Hello! What's your name?";
    cin >> name;
    cout << "How old are you?";
    cin >> age;
    cout << "\nHi " << name;
    cout << ",you are " << age << " years old" << endl;
    cout << "End of the program.." << endl;

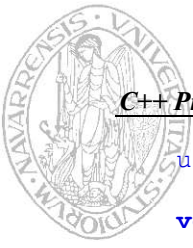
} /* end of main() */
```

You will have to create a new project named *Exer1* and the file containing the code, named *Exer1.cpp*. In order to compile and build the program you must use the **Build** menu and the menu options **Compile File** (for compiling), **Build** (for compiling and building) and **Crtl+F5** (for executing the program).

1.2 Exercise 2: A small arithmetic calculation using the Debugger

This program, in addition to calculating the distance of a *marathon* in kilometers, serves to learn to find errors, using the debugger. Write the code and save it as *marathon.cpp*.

```
// File marathon.cpp
// A marathon is 26 miles and 385 yards long. A mile is 1760 yards long.
/* Calculate the distance of a marathon in kilometers */
#include <iostream>
```



```
using namespace std;    //for Visual Studio 2019 or superior

void main (void) {
    int miles, yards;
    float kilometers;

    miles = 26;
    yards = 385;
    /* conversion formula */
    kilometers = 1.609 * (miles + yards / 1760);
    // data output
    cout << "A marathon is " << kilometers << " kilometers long" << endl;
}
```

You will have to create a new Project named **Exer02**. The result you should obtain is 42.1859; however, it is not so following these steps. Where is the mistake? Use the *debugger* to find it.

1.3 Exercise 3: Converter Program: Euros to dollars and vice versa.

Now it's time to try to do an exercise without having the solution on hand, for which you first need try to think of how to solve the algorithm using pen and paper and afterwards implement it using C++.

Design a program, which allows you to convert some Euros to Dollars and vice versa.

You must follow the next steps:

- First, the program must **ask** you if you want to convert from **Euros to Dollars** or **Dollars to Euros**
- Then, the program must **ask** for the **rate** and **quantity** to convert and,
- Finally, the program must **show the result** of the conversion on the **screen**.

1.4 Exercise 4: Second-degree equation's solver.

Let us dig a little deeper into mathematical algorithms with C++.

The formula of the second-degree equation is:

$$ax^2 + bx + c = 0$$

First, calculate the Discriminant:

$$discr = b^2 - 4ac$$

If **discr > 0.0** the roots are real and different:

$$x_1 = \frac{-b + \sqrt{discr}}{2a} \quad x_2 = \frac{-b - \sqrt{discr}}{2a}$$

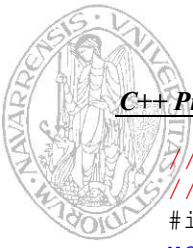
If **discr = 0.0**, the roots are real and equal:

$$x_1 = x_2 = \frac{-b}{2a}$$

Finally, if **discr < 0.0**, the roots are complex:

$$xr = \frac{-b}{2a} \quad xi_1 = \frac{+\sqrt{-discr}}{2a} \quad xi_2 = \frac{-\sqrt{-discr}}{2a}$$

Write down the program and save it as **Equat2.cpp**.



```
// Equation2
// Tecnun 2025 - PB
#include <iostream>
using namespace std;    //for Visual Studio 2019 or superior
#include <math.h>        // sqrt()
void main(void)
{
    double a, b, c;
    double discr, x1, x2, xr, xi;
    cout << "Enter the coefficients A, B and C: ";
    cin >> a >> b >> c;

    discr = ... ;
    if (discr > 0.0) {
        x1= . . . ;
        x2= . . . ;
        cout << "Real Roots: ";
        cout << x1 << " y " << x2 << endl;
    }
    else if (discr < 0.0) {
        xr= . . . ;
        xi= . . . ;
        cout << "Complex Roots: " << endl;
        cout << "(" << xr << ", " << xi << "i) y ";
        cout << "(" << xr << ", " << -xi << "i)" << endl;
    }
    else {
        x1= . . . ;
        cout << "Equal Roots: " << x1 << endl;
    }
}
```

Compile and run the program, trying with the next values:

A	B	C	Result
1	4	4	Equal Roots
1	5	4	Real Roots
1	3	6.25	Complex Roots

1.5 Exercise 5: Temperature Converter.

You must design a program to convert from $^{\circ}\text{C} \rightarrow ^{\circ}\text{F}$ and $^{\circ}\text{F} \rightarrow ^{\circ}\text{C}$. The formula for this is shown in the next equation:

$$^{\circ}\text{F} = \frac{9.0}{5.0} * (^{\circ}\text{C}) + 32$$

If you have not finished all the exercises, you must complete them at home.

Notes:

- Don't forget to add ";" always at the end of each statement.
- Don't forget to close the opened quotation marks, the parenthesis and the braces { }.
- Don't forget to declare the variables before using them.
- Finally, remember that C++ distinguishes between uppercase and lowercase characters.